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Transformative impact of pre-service CLIL science workshops on pedagogical content knowledge and language teaching competencies

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While pedagogical content knowledge (PCK) and The Language Triptych (i.e., Language of Learning, Language for Learning, and Language through Learning) provide new insights into Content and Language Integrated Learning (CLIL), little research exists integrating this knowledge into CLIL preservice teacher education in Taiwan. Grounded in Transformative Learning Theory, this study aimed to explore the impact of a CLIL science PCK and language teaching preparation workshop on preservice teachers' competencies in PCK and language teaching skills with a focus on the transformative potential of these interventions. Using a mixed-methods research design, the study involved 15 sophomore preservice CLIL science students from a Taiwanese university. Data were collected through pre- and post-workshop online microteaching sessions, written tests, and surveys, all of which assessed participants' PCK and language teaching competencies. The analysis of the pre- and post-workshop online microteaching sessions revealed significant improvements across all assessed criteria and suggested a positive transformation in participants' pedagogical and language teaching competencies. These changes mirrored important aspects of transformative learning with changes in pedagogies and content and language integration. According to the written tests, the participants made significant advancements in understanding CLIL methodologies as well as strategies for teaching language. Participants' evolving perspectives on CLIL science teaching and increased emotional awareness pointed to their growing preparedness for classroom implementation. Participants' responses in the survey reflected a noticeable shift in their perceptions of CLIL science teaching and emotional awareness, which highlights their increased readiness to adopt CLIL practices. The results show the training needs of CLIL preservice science teachers in Taiwan and offer practical strategies for designing effective training programs.

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Introduction

In order to promote linguistic diversity and train citizens for a globalized world, Content and Language Integrated Learning (CLIL) has been promoted by the European Union, which teaches subjects through additional languages other than the mother tongue (Pérez Cañado (2016)). CLIL is implemented in primary and secondary education in Taiwan in response to the demands of global academic and professional settings to improve English proficiency and integrate multilingual and multicultural competencies (Chen, et al., 2020; Tsou, 2018). However, Taiwan's CLIL implementation encounters significant challenges. The communicative and interactive nature of CLIL can be hampered by the traditional focus on rote learning and high-stakes testing that characterize primary and secondary education (Tsou, 2018). Moreover, given the recent implementation of CLIL in Taiwan, many schools are still exploring effective paths to use available resources and integrate globally recognized "best practices" into their school-based contexts (Yang, 2022). These factors make acquiring CLIL pedagogical and language teaching competencies for CLIL science teachers particularly challenging, especially given the complex nature of subjects like science (Sahin & Yilmaz, 2020).

It is evident that the lack of proper training for teachers impedes effective CLIL implementation in Taiwan. To overcome these issues, a successful CLIL teacher must have a very high level of proficiency in both the subject content and the target language and must also be skillful in blending language and content instruction. Nevertheless, existing teacher education programs in Taiwan commonly do not offer sufficient preparation in these domains (Pineda & Tsou, 2021), which is a considerable obstacle to the implementation of CLIL. Moreover, the lack of teachers who have the bilingual competence of the subject and the target language leads to ineffective collaboration between language and subject teachers (Luo, 2021). Preservice teachers should construct and utilize their knowledge, skills, and attitudes in their theoretical course in the teacher training programs (Karal Eyüboğlu, 2024); yet, they still cannot apply approaches to English teaching which impact positively on pupils' communicative competence, indicating that training modules need to be restructured (Kao, 2022). In balance, the tendencies in CLIL teacher education in Taiwan reveal the demand for tailored professional growth programs that can equip preservice teachers with the competencies and knowledge they need to effectively integrate language and content instruction in the classroom.

Integrating pedagogical content knowledge (PCK) with the Language Triptych into CLIL teacher training programs offers a promising solution to address the challenges faced in Taiwan's CLIL context. PCK, as conceptualized by Shulman (1987), is a special amalgam of content and pedagogy that enables teachers to organize and present subject matter in ways that are comprehensible to learners. This includes knowledge of instructional strategies, understanding of student learning processes, and the ability to interlink various domains of teaching, such as curriculum knowledge, classroom management, and assessment (Shulman, 1986). In the context of CLIL, language teaching skills are critical and involve the mastery of three interrelated types of language use, collectively conceptualized as the language triptych: language of learning, language for learning, and language through learning (Coyle et al., 2010). The "language of learning" involves subject-specific terminology for understanding content, the "language for learning" includes skills for classroom interactions and effective learning, and the "language through learning" encompasses incidental language acquisition through engaging with content (Coyle, et al., 2010). It is crucial for preservice CLIL science teachers to develop these skills since integrating language and content instruction allows for a richer, more interactive instructional environment. Such strong PCK while teaching language helps teachers to advance student understanding of critical scientific concepts and at the same time develop their language ability (Hudson et al., 2015).

Nevertheless, research studying the integration of PCK and the Language Triptych into CLIL preservice teacher education programs in Taiwan is limited. This study, grounded in transformative learning theory, aims to explore how participation in a 6-week CLIL science PCK and the language triptych preparation workshop could impact preservice teachers' competencies in PCK and language teaching skills. The impact was assessed using a mixed-methods approach, through pre- and post-workshop online microteaching sessions, written tests, and surveys among preservice CLIL science teachers. The research questions guiding this study are:

RQ1: How does participation in a six-week CLIL workshop influence preservice teachers' competencies in CLIL science pedagogical content knowledge and language teaching skills, particularly in areas such as content accuracy, clarity of explanations, student engagement, lesson planning, assessment strategies, scaffolding, and the use of the Language Triptych?

RQ2: What impact does the six-week CLIL workshop have on preservice teachers' understanding and application of CLIL science PCK and language teaching methodologies?

RQ3: What are preservice teachers' perceptions and experiences of transformative learning through the CLIL workshop, in relation to their views on CLIL science teaching, critical reflection, emotional awareness, social interaction, and preparedness for implementing CLIL practices?

Literature review

Theoretical framework: transformative learning theory. Transformative learning theory, initially formulated by Mezirow (1978), provides the theoretical framework for this study. Transformative learning theory forms the basis for adult learning through critical reflection and examination of the present old beliefs, assumptions, and values. This process is particularly relevant in educational settings where significant changes in perspective are required for effective teaching and learning (Mezirow, 1997). Transformative learning involves a series of stages that begin with a disorienting dilemma, an experience that challenges the learner's existing frame of reference. This is followed by self-examination and critical assessment of one's assumptions, leading to the exploration of new roles and identities. The process culminates in the reintegration of these new perspectives into the learner's life, resulting in a more inclusive, discriminating, and integrative worldview (Mezirow, 1978, 1991). In teacher education, transformative learning not only nurtures adaptability and reflective practices but also enables teachers to act upon various challenges introduced by the diversity of educational contexts (Arshavskaya & Reyes de la Paz, 2023).

The implementation of transformative learning theory demonstrates positive impacts on preservice teachers, according to empirical research. For example, Mangkhang et al. (2022) employed participatory action research to create transformative learning innovations for online teaching of social studies in Thailand. The analysis concluded that hybrid learning was well within the norm for higher education and was associated with improved teaching skills. Moreover, Birdsall (2022) explored the integration of environmental and sustainability education into teacher training in New Zealand, finding that reflective activities facilitated transformative learning for half of the participants. This suggests potential for the application of Mezirow's theory in sustainability-focused education, though its reach was limited to

certain participants. Similarly, Arshavskaya and Reyes de la Paz (2023) showed that another kind of reflective practice (i.e., writing autoethnographies) empowered teachers to reshape their teaching identities and improve classroom practices. Additionally, Esnati and Mukeredzi (2023) identified transformative learning experiences in Zimbabwe, finding that disorienting dilemmas and experiential activities were helpful to the transformations for some teachers. These studies collectively show that transformative learning, especially when it includes meaningful reflection and hands-on experience, can be a powerful way to help future teachers grow. At the same time, the varied outcomes suggest that transformation doesn't happen the same way for everyone. This points to a need for more thoughtful, tailored approaches that consider different educational contexts. If we want to make reflective practice a real part of preservice teacher training, it's important to create purposeful activities that not only challenge teachers to think differently, but also motivate them to take ownership of their learning.

However, the alternative research on transformative learning has largely not been adopted to advance CLIL science teacher education. Aiming to bridge this gap, the present study examines PCK and language teaching skills from a transformative learning perspective by exploring a series of workshops organized to enhance those knowledge components in Taiwan's CLIL contexts.

CLIL science teacher education. In CLIL science, educators deliver complex topics in scientific domain knowledge through a non-native language of instruction. Such a two-in-one approach is beneficial for both content mastery and language proficiency (Couto-Cantero & Bobadilla-Pérez, 2018; Kampen et al., 2018). CLIL science teachers play a crucial role in adapting to new educational and organizational conditions, ensuring that students acquire the necessary competencies in both content and language (Karal Eyüboğlu, 2024). The challenge of teaching science in a CLIL framework is compounded by the subject's inherent complexity, covering fields such as biology, earth science, chemistry, and physics (Sahin & Yilmaz, 2020). Despite the growing demand for CLIL science teachers and the evident benefits of such an educational approach, the professional development and certification of these teachers often remain neglected (Macaro, Han (2019)). Thus, the dynamicity of the characteristics of scientific domains and linguistic difficulties in CLIL environments require different teacher training programs (Sahin & Yilmaz, 2020) and necessitate a balanced integration of content knowledge and language teaching skills to meet global educational standards and student needs (Dalton-Puffer, 2011).

There is growing evidence that the integration of CLIL methodologies into science teacher education could be a promising path to evoke improved teaching practices in different contexts. In a qualitative study carried out at Volgograd State Medical University, Kovrizhnykh (2022) studied science teachers who were trained to teach physics through English as an interim language. Despite the diversity of practices within CLIL education to accommodate different language and content-related competencies through integrated linguistic approaches, the need for specific integrated linguistic techniques was emphasized clearly through the diversity in students' preparedness and differing methods to optimally influence language and content-related competencies. Likewise, Piacentini, Vieira, and Simões (2022) investigated the effects of integrating English into a science subject in a school located in Portugal. Their case study found that adding English brought science teachers' attention to students' language challenges, and they adapted their teaching strategies and resource use to support both language development and deeper learning in the science domain. In a further study, Nieto Moreno de Diezmas et al., (2024) explored the motivational effect of CLIL-izing english medium instruction (EMI) on preservice teachers' science education in Spain. The findings from their longitudinal study revealed a significant increase in motivation to learn science content over time, but no increase in motivation to acquire the L2 (English). Overall, the studies have all echoed the potential of CLIL science teacher education to improve the learning outcomes for content and language while also tackling some major concerns, such as students succumbing to linguistic obstacles in subject knowledge acquisition and heterogeneous preparedness for subject matter (science).

However, important gaps are still unresolved, including the integration of transformative learning frameworks in CLIL science teacher education toward supporting teachers' professional development relative to adapting teaching practice for diverse students. This study seeks to fill these gaps by deriving transformative, context-specific CLIL-based interventions in terms of both pedagogical content knowledge and linguistic scaffolding for science teacher education in Taiwan.

CLIL in Taiwan: challenges and transformative approaches in science teacher education. Policies such as the 2030 Bilingual Nation Policy, which plans to integrate English into the medium of instruction across many non-language courses, have been a driving force behind the rapid growth in Taiwanese bilingual education (Curran & Chern, 2021; Huang & Tsou, 2023). Within this initiative, CLIL has emerged as a prominent approach to blend language learning with subject content mastery to enhance global competitiveness (Chen & Tsou, 2021). CLIL aligns closely with Taiwan's Twelve-Year Basic Education Curriculum Guidelines, which emphasize cross-disciplinary learning, real-world application, and core competencies (Huang & Tsou, 2023). Despite the potential of CLIL, Taiwan's education system faces significant challenges in its implementation, particularly in preparing teachers for bilingual classrooms and addressing the linguistic and pedagogical complexities associated with CLIL science instruction (Chen & Tsou, 2021; Graham & Yeh, 2023).

Based on the evaluation of the empirical evidence presented, implementing CLIL in Taiwan provides both promise and challenges. As an example, Curran and Chern (2021) investigated the perspectives of in-service science teachers in summer science camps where preservice English teachers offered science-English CLIL courses. The teachers reported that the lessons were interesting and helped students in the camps use English as a medium of content learning. Nonetheless, it remained clear that significant scaffolding was required for preservice teachers and their students, again highlighting the importance of interdisciplinary crossover. While Chen and Tsou (2021) studied the practices of trans-semiotizing and translanguaging by local teachers in primary CLIL classrooms, these practices supported their strategies for bridging language disparities and promoting a positive learning atmosphere. Such practices also provided a space for teachers to intervene in both the politics of nativeness and the pressure to teach within native-speaker norms, while positively impacting language learning through cognitive benefits from content delivery-related practice and increased learner engagement. Furthermore, Graham and Yeh (2023) find that deliberate ambivalence toward bilingual education in official policy led to dramatically variable enactment across sites. They have noted that there are numerous inconsistencies in the language curriculum since these stem from the different background knowledge of teachers as to how much content is linked with the number of languages learned and, consequently, an exclusive focus on teacher training programs. Overall, the results of these studies suggest that CLIL can offer the promise to be able

to improve bilingual education in Taiwan by incorporating the learning material and learning a language through contextualized means.

Nonetheless, key gaps have been left unmet. For example, Curran and Chern (2021) did not touch upon the ways in which CLIL has an impact on students' later outcomes or how the blending of two disciplines can be maintained outside a specialist classroom. The effect of translanguaging on the cognitive and academic performance of students was not carried out in any other study except Chen and Tsou (2021), who discussed the significant role of translanguaging back references. Graham and Yeh (2023) find policy ambiguities without completing an analysis and then offer no speculation as to the degree this successfully meets a goal, or even what balance teachers reach for. These constraints imply that there is a pressing need for transformative, situated innovations that integrate PCK and language scaffolding in CLIL science teacher preparation. Given these constraints, the present study employs tenets of transformative learning frameworks to examine how preservice CLIL science teachers in Taiwan are prepared for Chinese-English bilingual instruction.

Pedagogical content knowledge in preservice teacher education. As conceived by Shulman (1987), PCK is a core framework for teachers, linking together the different components of content knowledge, pedagogical knowledge, curricular knowledge, learners, and context. PCK highlights that teachers need to comprehend the subject matter and teach it effectively by connecting theoretical with practical knowledge (Shulman, 1986). With regard to CLIL preservice teacher education, PCK is concerned with the ability to maintain classrooms, use effective teaching approaches, and foster a supportive learning environment (Richardson, 1996). This latter aspect includes the initiation and facilitation of classroom interactions that develop the language needed for a specific learning goal (language for learning), as well as fostering incidental outcomes in language development through content engagement (language through learning); collectively termed pedagogical content knowledge (PCK) for CLIL science teachers (Coyle, Hood, & Marsh, 2010). The integration of PCK into CLIL preservice teacher education is crucial because it helps future teachers develop a comprehensive understanding of how to teach complex scientific concepts in a second language, thereby enhancing their instructional effectiveness and self-efficacy (Hudson et al., 2015; Stohlmann et al., 2012).

Its development among preservice teachers has been explored in empirical studies, and different approaches focusing on diverse contexts to enhance preservice teacher PCK have been examined. In a qualitative case study in Turkey, Nalbantoglu et al. (2023) embedded lesson study into a science teacher education program at the middle school level. The cycle of lesson study (i.e., planning, teaching, reflecting, re-teaching, and re-reflecting) played a significant role in participants' improved understanding of curriculum, learners, instructional strategies, and assessment. In a similar manner, Botes (2024) examined the use of educational science board games to enhance preservice natural science teachers' PCK in South Africa. Using focus group discussions and photo-voice methodology, the study demonstrated that preservice teachers gained deeper insights into curriculum planning, instructional strategies, and learner engagement, highlighting the potential of hands-on, innovative tasks to develop pedagogical competencies. Furthermore, Çavaş, Kara, Anagün, and Ayar (2024) examined the mediating role of STEM attitudes in the relationship between STEM PCK and intra-class practice self-efficacy among 345 teacher candidates in Turkey. Their quantitative findings showed that STEM attitudes significantly mediated this relationship, emphasizing the importance of attitudinal factors in developing preservice teachers' confidence in implementing STEM practices. While these studies collectively affirm the effectiveness of targeted interventions in developing PCK, they reveal the need to explore its integration with language teaching skills in CLIL contexts, especially in science education.

Current research offers useful information about how PCK can be improved and what methods are effective, but it also shows gaps. For example, Nalbantoglu et al. (2023) and Botes (2024) were unable to identify innovative aspects such as lesson study and educational board games. Neither examines the dynamic relationships between PCK domains (e.g., engaging in identifying links between instructional practice with contextual knowledge), particularly in linguistically rich classrooms. Similarly, while Çavaş et al. (2024) delve into beliefs and highlight the contribution of attitudes in boosting self-efficacy, they do not investigate how preservice teachers enact PCK in classroom settings. These gaps highlight the need for a deeper understanding of how PCK can exist in and be utilized to support learning in CLIL contexts, where both language and content must be taken into consideration. Drawing upon these insights, this study seeks to address these limitations by investigating how workshops situated within transformative learning frameworks can align PCK with language teaching competencies. The findings will inform our understanding of the necessary skills and competencies to be developed among preservice CLIL science teachers in Taiwan, helping them equip themselves with effective instructional strategies and a thorough grasp of contextual knowledge to foster learner engagement.

Integrating the language triptych in CLIL education. Coyle et al. (2010) introduced the language triptych, a pedagogical framework intended to represent the more complex role language plays in CLIL contexts. It distinguishes among language of learning, language for learning, and language through learning in bilingual classrooms and shows their interrelatedness produced through knowledge construction and meaning-making. In line with the 4Cs of CLIL (Content, Cognition, Communication, and Culture), it also provides a structured approach for planning, teaching, and reflective practice in understanding how language works in this type of context. Its utility is evidenced in a variety of educational settings from K-12 to teacher-education programs (Coyle & Meyer, 2021). However, the work of integrating the Language Triptych into classroom practice is difficult due to the mixed linguistic and cognitive demands set upon learners (Dalton-Puffer, 2016).

The Language Triptych has been an area of study and interest in various studies investigating teaching and learning in CLIL contexts. For instance, Turner and Fielding (2021) examined how primary and secondary teachers in Australia applied the Language Triptych. They found that the framework expanded teachers' ability to address learners' linguistic needs, particularly through spontaneous moments of language through learning, although challenges in supporting meta-cognitive language functions were noted. Similarly, the Language Triptych was used as part of a study in which preservice teachers designed and implemented a language-focused CLIL lesson in Kao (2022) in Taiwan. Results showed that the teaching of translanguaging fostered language awareness and the creation of specific linguistic competencies, and an increasing recognition for the use of translanguaging as a means to scaffold learning. As part of an action research group with secondary teachers in Argentina, Banegas et al. (2025) studied the use of the Language Triptych. This language provided a scaffold for teachers as they tried to

Table 1 Content and structure of the 6-week CLIL science workshop for preservice teachers.

Week	Content and techniques	Rationale
Pre	Introduction to CLIL and objectives: Overview of CLIL principles, workshop goals, pre-tests, microteaching	Establishes baseline understanding; identifies participants' starting competencies (Kovrizhnykh, 2022).
1	CLIL science lesson planning: Principles of CLIL; step-by-step lesson planning for scientific topics	Addresses the lack of systematic lesson planning in CLIL contexts (Dalton-Puffer, 2011).
2	Teaching pedagogies & engagement: Strategies for active learning; practice integrating PCK and CLIL	Promotes effective teaching techniques and student engagement (Sahin & Yilmaz, 2020).
3	Language teaching and scaffolding: Focus on the language triptych; practical application through tasks	Integrates language-specific strategies critical for CLIL science teachers (Coyle et al., 2010).
4	Assessment and feedback: Types of assessments in CLIL; practice giving constructive feedback, post-tests.	Develops formative assessment skills critical for addressing linguistic and cognitive diversity (Chen & Tsou, 2021).
Post	Microteaching and reflection: Participants implement and receive feedback on CLIL teaching strategies	Encourages reflective practice, a key element of transformative learning (Birdsall, 2022).

meet the challenges of teaching academic integrity in a comprehensive way and ensure that students could be prepared through the use of interdisciplinary language. The study identified a Triptych and concurred that additional terms to describe learning, metacognitive, and reflective language learners needed to explain their learning was necessary, expanding on to proclaim an import from "Language Quadriptych." However, it also made clear how teachers indeed prioritized language of learning (the terminology that they used regularly to refer to content) and thus emphasized the vocabulary side more, detracting from emergent and transactional language. The results suggest the need for focused teacher training to successfully manage linguistic and cognitive complexities in CLIL classrooms.

In general, the studies conclude that the Language Triptych are beneficial for supporting effective CLIL instruction. However, several gaps remain. For instance, less is known about how teachers implement the language triptych in its entirety by providing language support systematically through learning and also opportunistically exploiting (noticing) learners' developing language needs during spontaneous discourse. Similarly, the studies handling applications in primary-level CLIL cases are few and far between to be found in bilingual education systems such as Taiwan. Based on the above, this present paper seeks to contribute to addressing an apparent gap in the current research and looks at how the three sides of the language triptych interact linguistically within CLIL classrooms in Taiwan. In particular, it discusses how the framework can promote scientific literacy and vocabulary growth among learners and considers the specific challenges that a bilingual education setting brings to practice in Taiwan. By expanding the language triptych to include metacognitive scaffolding strategies, the study seeks to add to knowledge about how CLIL pedagogies can be developed and implemented in teaching practice across different educational contexts.

Methods

Research design and participants. The study employed a mixed-methods research design, offering detailed insights into specific instances and broader principles (Cohen, Manion, & Morrison, 2007). The purpose of this research was to describe the phenomenon from a detailed examination of how preservice CLIL science teachers experienced it after presenting the workshop (Yin, 2003). Purposive sampling (Patton, 2002) was used, and the participants were 15 sophomore preservice CLIL science students, aged 19 to 20, from the Department of Science Education at a university in Taiwan. The participants included six males and nine females, all native Mandarin Chinese speakers with an intermediate level of English proficiency (defined by their academic records and a pre-test). None of them had previous experience in pedagogical or content knowledge courses for teachers, nor had any of them taught science in a CLIL context prior to the 6-week CLIL science workshops. Ethical considerations stated that participation was voluntary, informed consent was guaranteed, and confidentiality and anonymity were maintained.

Workshop content and structure. The 6-week CLIL workshop was conceptualized as a powerful intervention to transform the PCK and language teaching competencies of preservice teachers. Based on Transformative learning theory (Mezirow, 1978; 1997), the workshop focused on integrating content and language in a content-subject context of science education for CLIL practice in Taiwan. Utilizing the Language Triptych framework (Coyle et al., 2010), the workshop sequentially introduced participants to how to select and use language of learning, language for learning, and language through learning in lesson planning and classroom discourse. This fosters the practical applications in their teaching.

The workshop was conducted by an experienced, near-native-like CLIL science teacher from Taiwan with a strong theoretical background in CLIL and English pedagogy, as well as 5 years of experience teaching CLIL science in elementary schools. To facilitate experiential learning, the instructor used a CLIL science textbook alongside PowerPoint presentations, worksheets, and activities drawn from the 5 years of CLIL science teaching experience. Each session had a specific focus, such as lesson planning, scaffolding, the language triptych, student engagement, or assessment, and the instructor used one textbook unit per session to demonstrate strategies targeting that session's theme. Treating the participants as elementary school students, the instructor modeled teaching step by step, scaffolding participants' understanding and skills throughout the workshop. This immersive, practice-oriented approach helped participants build thematic awareness and develop their ability to enact key dimensions of CLIL teaching. The weekly progression of the workshop is summarized in Table 1.

Unique features of the workshop design. The workshop was restructured to relate theory-to-practice through the instructor's demonstration, hands-on learning, and microteaching. The specific features of the CLIL science education theory-to-practice bridge in the context of the workshop are summarized in points as follows:

1. Systematic incorporation of the language triptych
   Through demonstrations and guided discussions, the instructor methodically introduced participants to the components of the language triptych. All parts were further reinforced through specific exercises and group practice.

- Language of learning (subject-specific terminology): In a lesson on "air composition," the instructor highlighted key terms such as oxygen, carbon dioxide, and nitrogen. Visual aids, bilingual annotations, and contextual examples were used to help participants understand and teach these terms effectively.
- Language for learning (classroom interactions): The instructor modeled procedural dialogs, such as, "What happens when we remove air from the jar?" Participants practiced crafting and using similar dialogs in group activities to promote classroom interaction and procedural discourse.
- Language through learning (incidental acquisition): While carrying out a practical task on designing succulent pots, participants verbalized in English what their groups were doing while planting. They discussed the process and selected the correct materials using vocabulary like roots, soil, and water drainage. The task-intrinsic orientation drew on incidental vocabulary learning by having participants engage in meaningful, context-based exercises.

2. Reflective activities
   Reflective practices were integrated during or after each session. After participants demonstrated how they might teach certain scientific concepts or terminology, the instructor provided reflective feedback, offering constructive insights into their teaching methods, language use, and engagement strategies. Additionally, after participants micro-taught, they received guided feedback, based on a rubric. For instance, in the feedback after one participant's lesson on the states of matter, a suggestion on incorporating student engagement strategies was given to the participant. Participants adapted their teaching and plans based on feedback, including the phases of transformative learning (Mezirow, 1997), such as self-examination and reestablishing themselves with perspectives gained.
3. Hands-on, Integrated activities
   Every session integrated scientific knowledge, language training, and practical activities. For instance, participants practiced how to teach a lesson on air composition by creating a pie chart that showed the relative proportions of gases in the atmosphere. Not only did this activity conceptually strengthen vocabulary (i.e., percent, composition, chart), but it also engaged students on a more meaningful level with the scientific concepts. Reflecting the above, these activities demonstrate why scaffolding is beneficial for unifying quantitative language proficiency and scientific inquiry (Piacentini et al., 2022).

Workshop teaching techniques and rationale.

1. Scaffolding instruction
   The instructor scaffolded each lesson by starting with warm-up questions, activities, or videos that introduced small, manageable chunks of information, gradually progressing to the more complex scientific concepts of the unit. For instance, during a lesson on the water cycle, students were introduced to terms such as evaporation and condensation with visual aids and simple comparisons. The instructor then prompted participants to link these terms to more complex concepts, like the impacts of temperature and pressure, with iterative questions like, "What occurs when water vapor cools?" This approach closed content and language gaps, strengthening participants' comprehension and instruction abilities.
2. Interactive learning strategies
   To encourage participation, the instructor used strategies such as group discussions, think-pair-share, and role-playing:
   - Using sentence starters, students discussed in groups the importance of oxygen to living organisms ("Oxygen is important because…")
   - Think-pair-share: Working in pairs, students responded to questions like "What would happen if there was no oxygen?" before they spoke in front of the class.
   - Role-playing: The participants role-played the force of gravity and an object falling. For instance, one participant, acting as gravity, "pulled" a second participant, acting as a falling object, toward the ground, with English-language explanations such as "Gravity pulls objects towards the Earth." These personas cement the vocabulary associated with gravity and make the abstract idea behind it much more tangible.
3. Task-based language teaching (TBLT)
   The instructor demonstrated examples of fun activities an elementary-schooler might engage in, like building bird nests or playing VR games designed to teach subjects like force and gravity. Participants in one session wore VR headsets and played VR games to investigate scientific concepts such as forces and gravity through problem-solving tasks. These were the types of hands-on activities that integrated scientific inquiry with language, making abstract concepts more concrete.
4. Translanguaging practices
   Mandarin and English were encouraged to be strategically incorporated in CLIL science PowerPoint slides to support bilingual learning as advocated by scholars (e.g., Chen & Tsou, 2021). For example, PowerPoint slides on friction included bilingual terms such as friction (摩擦力). Although scientific concepts were conveyed in English, some technical terms were reinforced in Mandarin Chinese. Students brainstormed ideas in Mandarin and wrote answers, but then presented their answers in English, a technique that allowed students to construct understanding in their first language before transitioning into their target language.

Data collection and analysis

Pre- and post-workshop online microteaching sessions. Microteaching sessions were carried out online before and after the CLIL workshop to evaluate the workshop's effect on participants' PCK and language teaching abilities. This is to create a low-pressure environment for the participants as they only teach two pro-CLIL science experts. This setup reduced performance pressure and allowed subjects to concentrate on both building and displaying their pedagogical skillset.

To establish a baseline, participants chose a unit from the CLIL science textbook and micro-taught it bilingually or in English for 3–5 minutes while explaining their strategies prior to the workshop commencement. One week after completing the workshop, participants conducted a second session to demonstrate their integration of CLIL principles. To ensure systematic and fair evaluation, the sessions were assessed using validated rubrics (Appendix A) developed based on Shulman's (1986) PCK framework and Coyle et al. 2010 language triptych. These rubrics were tailored to address the specific dimensions of PCK and language teaching, including content accuracy, clarity of explanations, student engagement, lesson planning, assessment

Table 2 Wilcoxon signed-rank test results for pre- and post-workshop microteaching sessions.

Criterion ($n = 15$)	Pre-workshop median (IQR)	Post-workshop median (IQR)	Z	p value	Effect size (r)
Pedagogical content knowledge (PCK)					
Accuracy of content	1 (1-2)	3 (3-3)	−3.497	0.000*	0.903
Clarity of explanations	1 (0-1)	3 (3-4)	−3.330	0.001*	0.860
Engagement of students	1 (1-2)	3 (3-4)	−3.497	0.000*	0.903
Lesson planning	1 (1-2)	4 (3-4)	−3.496	0.000*	0.903
Assessment techniques	1 (1-1)	3 (3-4)	−3.438	0.001*	0.888
Implementation of scaffolding techniques	1 (0-1)	3 (3-4)	−3.453	0.001*	0.892
Language teaching skills					
Use of language of learning	1 (1-2)	4 (4-4)	−3.462	0.001*	0.894
Use of language for learning	0 (0-1)	3 (3-4)	−3.475	0.001*	0.897
Use of language through learning	0 (0-1)	3 (3-4)	−3.499	0.000*	0.903

The rubrics were used by two raters based on a five-point scale.
IQR interquartile range.
*indicates significance at $p < 0.003125$.

techniques, scaffolding, and the three components of the language triptych.

Each dimension was rated on a five-point scale, ranging from "no evidence" (0 points) to "advanced understanding" (four points), with detailed descriptors provided to guide evaluations. The rubrics were piloted with two experienced CLIL science educators, who tested them on three mock microteaching sessions to ensure their validity and reliability in assessing preservice teachers' performances. High interrater reliability (ranging from 0.85 to 1.0) was achieved through calibration sessions, during which raters aligned their scoring criteria.

The data were analyzed for normality using the Shapiro–Wilk test, which indicated non-normal distributions, requiring the use of nonparametric methods. Slight deviations from normality were confirmed by descriptive statistics: skewness and kurtosis. For the comparison between pre- and post-workshop scores, the Wilcoxon signed-rank test was used (with Bonferroni correction of the alpha level [0.003125] for multiple comparisons across 16 evaluation criteria).

Pre- and post-workshop CLIL PCK and language teaching written tests. Pre- and post-test written evaluations of participants' knowledge of CLIL PCK and theories of language teaching were developed and administered. The two CLIL science professionals who assessed the microteaching sessions also designed and validated the tests, which provided consistency with the assessment framework. The test items were adapted to the Taiwanese CLIL science education context, which incorporated 20 multiple-choice and ten true/false questions.

The multiple-choice questions focused on PCK components such as scaffolding, student engagement strategies, and assessment techniques, while the true/false questions evaluated participants' knowledge of the language triptych and its practical applications. Test items were also reviewed by a panel of two experts in CLIL science education to enhance test validity and ensure test item clarity, relevance, and alignment with workshop content. Revisions of the tests were made, and the tests were pilot tested with another group of similar preservice teachers to ensure that the questions were appropriately challenging and understandable. The pre- and post-workshop tests were matched for difficulty, with the post-workshop test reworded to minimize memory effects. Cronbach's Alpha confirmed internal consistency with scores of 0.867 for the pre-test and 0.846 for the post-test. The reliability of all subscales was strong, with PCK-related items producing 0.827 (pre) and 0.839 (post); language teaching items, 0.719 (pre) and 0.743 (post). The test score distributions approximated normality as confirmed by the Shapiro–Wilk test (pre: $W = 0.925$, $p = 0.232$; post: $W = 0.944$, $p = 0.429$), as well as skewness and kurtosis values. To compare pre- and post-test scores, the Wilcoxon signed-rank test was performed, with the alpha adjusted for multiple comparisons ($\alpha > 0.003125$, Bonferroni-corrected).

Pre- and post-workshop surveys and analysis. Pre- and post-workshop surveys (Appendix B) evaluated participants' readiness and effectiveness in CLIL science teaching. Developed based on transformative learning theory (Mezirow, 1997), these bilingual surveys (in English and Mandarin Chinese) covered perspectives in five areas on CLIL science teaching, engagement in critical reflection, emotional awareness and management, the value of social interactions, and preparedness for new CLIL teaching practices. Participants rated their agreement on a 5-point Likert scale. The pre- and post-workshop surveys showed high internal consistency, with Cronbach's Alpha values of 0.908 (pre) and 0.945 (post), respectively. The Shapiro–Wilk test showed most survey items significantly deviated from a normal distribution (p values <0.05). Descriptive statistics revealed a slight positive skew (0.499, SE = 0.580) and platykurtic distribution (−0.736, SE = 1.121) in the pre-workshop survey, while the post-workshop survey had a slight negative skew (−0.433, SE = 0.580) and a distribution closer to normal (−0.419, $SE = 1.121$). Due to the small sample size, nonparametric tests were necessary; thus, Wilcoxon signed-rank tests were employed to identify significant improvements across the five areas, with an adjusted alpha level of 0.003125.

Data collection procedure. Data collection was conducted throughout the six-week workshop to evaluate the impact on participants' PCK and language teaching skills. Before the workshop, participants completed a written test and survey, followed by pre-workshop online microteaching recordings. In the final week, participants took the post-workshop written test and survey, and recorded their post-workshop microteaching sessions. These data sources, including lesson plans, written tests, and surveys, were used to assess the effectiveness of the workshop in enhancing preservice teachers' competencies.

Results

Pre- and post-workshop online microteaching (RQ1). Wilcoxon signed-rank tests (all p values <0.003125, adjusted using Bonferroni correction) confirmed that there was significant improvement across all criteria measured in the online microteaching sessions conducted post-workshop compared to pre-workshop. These statistics indicate the degree to which the workshop affected PCK and language teaching skills (Table 2).

Table 3 Wilcoxon signed-rank test results for pre- and post-workshop written tests.

Criterion ($n = 15$)	Pre-workshop median (IQR)	Post-workshop median (IQR)	Z	p value	Effect size (r)
Understanding of CLIL PCK ($i = 21$)	8 (4–10)	17 (16–19)	−3.3	0.001*	0.852
Language teaching methodologies ($i = 7$)	2 (1–3)	5 (4–5)	−3.099	0.002*	0.800

Significance of $p < 0.003125$ is indicated by an asterisk (*).
IQR interquartile range, *i* item.

Table 4 Wilcoxon signed-rank test results for pre- and post-workshop surveys.

Criterion ($n = 15$)	Pre-workshop median	Post-workshop median	Z	p value
Perspectives on CLIL science teaching (Items #1–3)	10.0	13.0	−3.104	0.002*
Engagement in critical reflection (Items #4–6)	12.0	13.0	−1.676	0.094
Emotional awareness and management (Items #7–9)	10.0	11.0	−2.215	0.027*
Value of social interactions (Items #10–12)	13.0	13.0	−1.169	0.243
Preparedness for new CLIL teaching practices (Items #13–15)	12.0	13.0	−2.408	0.016*

Significance of $p < 0.003125$ is indicated by an asterisk (*); participants rated their agreement with statements on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

For PCK, participants showed large improvements in their accuracy of content (pre: 1, IQR 1-2; post: 3, IQR 3–3), clarity of explanations (pre: 1, IQR 0–1; post: 3, IQR 3–4), and engagement of students (pre: 1, IQR 1–2; post: 3, IQR 3–4). With regard to post-workshop skill improvement, areas such as (1) lesson planning (pre: 1, IQR 1–2; post: 4, IQR 3–4) and (2) assessment tools (pre: 1, IQR 1–1; post: 3, IQR 3–4) also showed significant improvements. There were also significant improvements in the language of learning, language for learning, and language through learning in relation to language teaching skills. Their competencies in integrating language teaching strategies in the sciences were enhanced (all criteria pre: 0–1, post: 3–4). This demonstrates that the workshop effectively prepared participants to deliver scientific content in English, which in turn increased the efficiency of language acquisition. In conclusion, the overall effect sizes (from 0.860 to 0.903), categorized as moderate to large effects, indicated that the workshops were effective in contributing to the PCK and the language teaching components of preservice CLIL science teachers.

Pre- and post-workshop written tests (RQ2). The analysis of the pre- and post-workshop written tests revealed significant improvements in the participants' understanding of both CLIL PCK and language teaching methodologies (Table 3). The significance level for these results was adjusted using a Bonferroni correction, setting the alpha level at 0.003125, to account for multiple comparisons. For Understanding of CLIL PCK, the median score increased from 8 (IQR = 4–10) in the pre-workshop assessment to 17 (IQR = 16–19) post-workshop. The Wilcoxon signed-rank test results showed a Z-value of −3.3 and a p value of 0.001, indicating a highly significant enhancement in participants' PCK. The effect size for this change was 0.852, demonstrating a large impact of the workshop on participants' PCK. In the language teaching methodologies category, participants' median scores rose from 2 (IQR = 1–3) pre-workshop to 5 (IQR = 4–5) post-workshop. The Z-value of −3.099 and p value of 0.002 reflect significant improvements in participants' understanding and application of effective language teaching strategies within the CLIL framework. The effect size for this improvement was 0.800, also indicating a substantial impact. These findings reveal the workshop's effectiveness in enhancing both specific content knowledge and broader language teaching competencies among preservice teachers.

Pre- and post-workshop surveys. The analysis of the pre- and post-workshop surveys showed mixed results regarding significant changes in participants' perceptions and experiences (Table 4). For the perspectives on CLIL science teaching category, the median score increased from 10.0 pre-workshop to 13.0 post-workshop. The Wilcoxon signed-rank test yielded a Z value of −3.104 and a p value of 0.002, indicating a statistically significant improvement in participants' views on CLIL science teaching, with the change being significant even after applying the Bonferroni correction. In the Emotional Awareness and Management category, the median score rose from 10.0 to 11.0, with a Z value of −2.215 and a p value of 0.027, indicating significant gains in this area. While engagement in critical reflection and value of social interactions did not change significantly ($p = 0.094$ and 0.243, respectively), for the preparedness for new CLIL teaching practices, there was significant improvement; the median of the score progressed from 12.0 to 13.0 ($Z = -2.408$; $p = 0.016$).

Discussion

This study highlights that the CLIL workshop proved to be effective in developing preservice teachers' pedagogical content knowledge (PCK) and language teaching skills. These noticeable improvements in participants' ability to integrate content and language instruction reflect the theoretical underpinnings of transformative learning theory (Mezirow, 1978; 1997) and the language triptych framework (Coyle et al., 2010) as delineated in the Literature Review. These guidelines informed creative CLIL teaching and learning designs, both theoretically and practically, as they encompassed central pedagogical considerations, such as the intertwining of the processes of experiential learning and the formation of reflective learners, as well as the interrelationship between content and language in CLIL. These principles shaped the design of this workshop, allowing for a transformative experience that fit the participants' specific needs and context.

Enhanced PCK and teaching skills: addressing complex scientific concepts. These improvements in participants' ability to accurately explain scientific concepts, engage students, and design scaffolded lessons are a function of the workshop's intentional integration of Shulman's (1987) conceptualization of PCK. Through the CLIL science textbook prepared by the workshop instructor, coupled with an experiential practice and methodological modeling by the instructor, participants were provided

with strategies to implement in the field in order to access these abstract scientific perspectives. Building bird nests and playing VR games, for instance, led them to realize how interactive activities allow the linking of theoretical concepts like force and gravity with experiential learning experiences. This is in accordance with Sahin and Yilmaz (2020), who claimed the complexity within science should not be ignored in any CLIL context, as it should be tackled and approached innovatively.

Additionally, the scaffolding techniques implemented in the workshop align with some of the approaches described by Nalbantoglu et al. (2023), who found that the facilitated, discovery-based walkthroughs helped rewire lessons for participants in a way that made sense to them. The participants demonstrated their advancing instructional design capabilities by using scaffolded questioning to navigate complex concepts. Rather than focusing solely on singular terms (e.g., evaporation and condensation), they developed more comprehensive explanations of phenomena (e.g., the interplay of temperature and pressure). This approach is consistent with the Literature Review, which stresses the importance of systematic teacher training in order to develop effective PCK in CLIL environments (Dalton-Puffer, 2011; Hudson et al., 2015).

Language integration: the role of the language triptych. The focus on language of learning, language for learning, and language through learning allowed workshop participants to take away a solid framework for integrating language into their practice. The intentional integration of the language triptych enhanced participants' understanding not only in curriculum or subject-specific vocabulary but also their ability or capability to use the language itself to scaffold classroom situations and enable serendipitous language development. However, designing pots for succulents invites native usage of relevant vocabulary, serving incidental vocabulary learning (e.g., roots, soil, water drainage). Such results are consistent with the studies of Turner and Fielding (2021) discussing the phenomenon of the spontaneous use of language in CLIL contexts and respond to the mentioned gaps in the Literature Review related to the systematic adoption of the language triptych in teacher education.

Reflective practices: a foundation for transformative learning. The transformations of the workshop were embedded in reflective activities like the instructor's feedback to the students' in-class practical teaching and microteaching, along with its rubric-based feedback. Mezirow (1997), Esnati, and Mukeredzi (2023) contemplate critical reflection as the bedrock for professional development and transformative practice. Participants' ability to adapt their teaching plans based on feedback exemplifies the stages of transformative learning, including self-examination and the reintegration of new perspectives. However, the lack of significant improvement in Engagement in Critical Reflection observed in the survey results suggests that additional support may be needed to foster deeper reflective practices, particularly in educational contexts like Taiwan, which traditionally emphasize rote learning over critical inquiry (Tsou, 2018; Yang, 2022).

Bridging the gap in CLIL science teacher education. The workshop addressed concerns raised in the Literature Review, which had revealed gaps, especially regarding the transformation of CLIL science teachers in Taiwan, specifically contextualized training. The implementation of hands-on and collaborative activities, along with theoretical instruction, offered a blended approach to the workshop, which is in line with recommendations by Chen & Tsou (2021) and Graham & Yeh (2023). Moreover, the substantial improvement observed in participants' microteaching scores, combined with their more frequent use of English and implementation of practical demonstrations during sessions following the workshop, illustrates that the workshop was effective in equipping preservice teachers with the skills needed to navigate the linguistic and pedagogical hurdles of CLIL science instruction.

Challenges and implications for future workshops. Overall, the workshop was a success; however, social interaction was an area of improvement. The reflection in the workshop was that, although CLIL experts were taken into account in the process, the restricted time oriented on peer collaboration possibly prevented the participants from getting extenuated with the social process of learning. This finding aligns with Karal Eyüboğlu, 2024, who emphasized the importance of collaborative learning in teacher education. Future workshops should consider incorporating more group-based activities and peer feedback mechanisms to enhance participants' appreciation of social interactions as a vital component of professional growth.

Moreover, the observed challenges in fostering critical reflection suggest a need for more structured and guided reflective practices. As highlighted by Birdsall (2022), reflective activities should be tailored to the participants' prior experiences and cultural context to maximize their impact. Incorporating targeted strategies, such as reflective journaling or peer-led discussions, may help participants engage more deeply with their teaching practices and overcome the constraints of traditional educational paradigms.

Conclusion

The findings of this study demonstrate the transformative impact of a six-week CLIL science teacher preparation workshop on preservice teachers' PCK and language teaching competencies. This transformation was evident in the significant improvements observed in participants' microteaching sessions, written tests, and survey responses. Participants demonstrated marked progress in key areas such as scaffolding techniques, language integration, and lesson planning, which are central to effective CLIL instruction. These improvements align with Shulman's (1987) conceptualization of PCK, emphasizing the integration of subject-specific knowledge with pedagogical strategies, and support Coyle et al., 2010 language triptych framework for effective language instruction in bilingual contexts.

The study confirms that the workshop's design, combining theoretical components with hands-on activities and reflective practices, successfully bridged the gap between theory and practice in CLIL education. For example, the use of practical, interactive tasks like building bird nests or exploring gravity through VR games not only helped participants connect abstract scientific concepts with real-world applications but also facilitated incidental language learning (e.g., roots, soil, water drainage). These findings echo Sahin and Yilmaz's (2020) emphasis on addressing the inherent complexity of science education in CLIL contexts through innovative approaches and support Turner and Fielding's (2021) observation that the spontaneous use of language fosters deeper engagement in CLIL classrooms.

Furthermore, the reflective practices embedded in the workshop, such as rubric-based feedback after microteaching, played a pivotal role in participants' professional growth. As suggested by Mezirow (1997), critical reflection is foundational to transformative learning, and this study's results indicate that structured reflective activities enabled participants to self-examine and adapt their teaching strategies effectively. Nonetheless, the findings of the surveys showed little advancement in critical reflection and respect for social interaction, indicating cultural interferences in

Taiwan's education system, where rote memorization is emphasized (Tsou, 2018; Yang, 2022). Bridging these gaps will be crucial to future workshops' transformative potential.

Suggestions for CLIL science teacher preparation in Taiwan. Future improvement in preparing CLIL science teachers in Taiwan can adopt more structured peer collaborations and interactive learning in workshops. Although the current workshop focused on the individual-level practice, the addition of group-based activities and peer feedback mechanisms may have provided a deeper insight into collaborative learning as an important professional competency (Karal Eyüboğlu, 2024). Providing opportunities to get together for collaborative lesson planning or peer-led discussions are two examples of this type of outcome.

Moreover, incorporating a practicum element within the program could expose preservice teachers to practical classroom environments, allowing them to practice what they have learned in real classrooms. Real experience would put them in better stead in tackling issues such as classroom management, student engagement, and integrated content and language instruction.

This study also illustrated the necessity of psychological and emotional support, as participants reported increased emotional awareness and emotional management. We need training programs that feature resilience-building and ways of preparing the emotional ground with teachers so that they are equipped to not only deliver effective instruction but also restore self-efficacy and motivation for teaching in arduous classroom contexts (Esnati & Mukeredzi, 2023). Finally, adopting a longitudinal perspective on professional learning efforts allows researchers to examine the persistence of skills and knowledge gained through workshops. This approach can also reveal how preservice teachers apply these proficiencies in practice, the extent of their implementation after initial training, and how they adapt their competencies over time.

Limitations. Some limitations of this study must be acknowledged. The first is that the sample was small ($n = 15$), limiting the statistical generalizability of the results and the statistical power to detect smaller effects. Second, nonparametric statistical tests (e.g., Wilcoxon signed-rank test) may provide less sensitivity for detecting subtle differences or changes despite their adherence to the distributional properties of the data. Although the Bonferroni correction was applied to mitigate Type I errors, this stringent adjustment increases the risk of Type II errors, potentially overlooking significant findings. Third, the microteaching sessions were conducted online in front of two CLIL science professionals rather than in real classroom settings. This arrangement does not completely match the circumstances of classroom teaching, especially in terms of student conversation and classroom management. As such, the findings may not fully reflect the real-world challenges of teaching experienced in real-world contexts.

Suggestions for future studies. Future studies should attempt to recruit larger and more heterogeneous populations to strengthen the generalizability of the results. More diversified participant groups would enhance the statistical analyses and offer greater insight into how the workshop benefits various teacher groups. Also, future research can add real classroom practicum to content observed, which would require a more holistic assessment of teaching behaviors, such as management styles and how well students engage with both the lesson and the teacher. Finally, a longitudinal study design would be particularly valuable in examining the sustainability of the workshop's impact on preservice teachers' development. Such research could explore how the skills and knowledge acquired during the workshop translate into long-term teaching practices and influence student outcomes in bilingual science education. These directions would not only address the limitations of this study but also contribute to advancing the field of CLIL teacher education by providing a clearer roadmap for effective professional development.

Data availability

The datasets used and analyzed during the current study are available in the Zenodo repository, accessible via the following https://doi.org/10.5281/zenodo.14289157. (Note: To access the dataset, please enter "14289157" in the search bar on Zenodo.org.). This repository provides open access to all the raw and curated data.

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Author contributions

The authors conducted the research, analyzed the data, and prepared the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

Formal ethical approval was not sought for this study, as ethical review is not mandatory for classroom-based studies involving university students in Taiwan when the research poses minimal risk, as confirmed through correspondence with the Institutional Review Board (IRB) at National Chung Cheng University, Taiwan, on December 11, 2024. Nevertheless, all participants were adults aged 18 or older, and written informed consent was obtained prior to the study, ensuring their understanding of the research objectives, voluntary participation, anonymity, and the right to withdraw at any time. The study adhered to established ethical principles and received positive feedback from participants, reflecting a supportive and constructive research environment.

Informed consent

Written informed consent was obtained from all participants before the study began. The consent covered participation in the study, data collection, and publication of the research findings. The consent process was conducted by the authors on April, 10, 2023. All participants were informed that anonymity would be maintained, and participation was voluntary, with the option to withdraw at any time.

Additional information

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